Data Analytics Capstone

Reducing False Alarm Review in PCB Manufacturing Using Inspection Data

Interim Report

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Repository and Report Navigation

Project repository: https://github.com/sahasrm06/Capstone_Proposal

Original data: https://data.mendeley.com/datasets/99jzmh9658/2

This standalone interim report follows the approved October 12, 2026 synopsis. The main text meets the required 15–20 page range. Appendix A contains supporting tables and figures; Appendix B gives reproducibility and data access instructions; Appendix C displays equations; Appendix D maps the template sections. The repository provides analysis scripts, derived datasets, results, and source-access instructions.

The raw files match both source checksums. All reported analysis uses these verified observations. The GitHub Data Availability Statement identifies the analysis files and derived outputs in the repository and explains source-linked access to the original raw data.

Introduction

Background and Context

Printed circuit board (PCB) production commonly uses surface-mounted technology to assemble components before automated optical inspection (AOI). AOI flags measurements that fall outside configured acceptance limits. A subsequent manual inspection station determines which flagged observations are actual defects. This study addresses the capacity consumed when a flag is ultimately judged nondefective. The source publication describes measurements from a real Siemens production line, making the setting directly relevant to electronics manufacturing quality management (Pfab et al., 2024).

The operational stakeholders are inspection personnel, production supervisors, and quality engineers. Inspectors need useful priorities, supervisors need a manageable review queue, and quality engineers need an explicit account of defects that a proposed rule would miss. Reducing review volume without measuring the associated detection loss would therefore provide an incomplete business recommendation. The study evaluates the tradeoff retrospectively, using manually assigned labels as the available reference outcome.

This interim analysis establishes alarm-stream composition, configuration risk, temporal variation, and preliminary supervised performance. The selected histogram gradient boosting was chosen using training-day folds, with threshold selection restricted to validation days. The final rule captures 98.37% of positive test groups while identifying 11.47% of all test groups for candidate avoidance. These are retrospective group outcomes. Independent-group intervals clear both targets, but day-block avoidance uncertainty does not establish a reliable reduction above 10%. The operational recommendation remains manual review.

Problem Statement

The business problem is unnecessary manual review of AOI alarm groups that contain no confirmed defect. The analytical gap is whether inspection information already available

before manual review can identify a lower-risk portion of that queue while retaining more than 90% of confirmed-defect groups. A candidate rule must also identify more than 10% of all alarm groups for potential review avoidance. These two criteria jointly define the academic research objective; neither is an approved production release standard.

Purpose of the Study

The study's purpose is to characterize configuration risk, develop a supervised risk score, convert that score into a fixed review rule, and examine whether prevalence varies over time. Its primary dependent variable is group_defect, defined as the maximum manual class label among records sharing a timestamp. Inspection measurements, inspection type, and anonymized contemporaneous metadata are potential explanatory variables. The rule's retention indicator produces review avoidance as a derived outcome.

Timestamp groups are a board-grouping proxy because verified board identifiers are absent. Consequently, an avoided group is not automatically one avoided physical PCB review, and a reduction in group count is not a measured reduction in staff hours. The project makes this distinction explicit to keep the operational conclusion proportionate to the evidence. If review duration is later measured, candidate time capacity can be calculated as avoided groups multiplied by minutes per group. Current data do not support monetary return, warranty-cost, or factory-yield estimates.

Interim Project Status (Progress Snapshot)

- Completed: raw-file verification, cleaning audit, EDA, sample-size calculations, 14-candidate chronological comparison, frozen-rule evaluation, and dependence checks.
- In progress: final interpretation, template alignment, and submission preparation.
- Pending: independent reproduction, deduplicated-model retraining, further feature-semantic verification, and prospective validation.

Scope and Objectives

The study population comprises all published AOI-flagged records from one production line. It excludes products accepted directly by AOI. The retrospective scope includes tabular data auditing, timestamp grouping, inspection-configuration comparisons, chronological model development, and uncertainty estimation. It excludes image interpretation, causal diagnosis of soldering problems, deployment, and claims about a particular set-top box product.

The objectives are to compare inspection-configuration risk, test defect capture by a fixed review rule, quantify candidate review avoidance, and examine temporal prevalence variation. The four research questions are stated below.

Research Question 1 (RQ1)

Does the proportion of groups containing a confirmed defect differ among the five dominant inspection types?

Research Question 2 (RQ2)

For a rule fixed before final testing, does the retained review queue capture more than 90% of confirmed defect groups?

Research Question 3 (RQ3)

Does that rule identify more than 10% of all alarm groups as candidates to avoid review, subject to the RQ2 detection criterion?

Research Question 4 (RQ4)

Does confirmed defect prevalence among alarm groups differ between the first 50 and last 51 active production days?

Literature Survey

Literature Review Approach

The literature review combines manufacturing-specific evidence with statistical and machine-learning methods needed to evaluate an imbalanced, temporally ordered alarm stream. Sources were selected for direct relevance to false-call reduction, inspection-data provenance, validation bias, classification under imbalance, interval estimation, or sample planning. The starting bibliography is the approved synopsis; publisher and bibliographic records were checked for the source dataset article and selected evaluation studies. This is a targeted methodological review rather than an exhaustive systematic review. Search concepts were AOI false calls, imbalanced classification, temporal validation, and proportion uncertainty. PubMed/PMC and publisher records supported verification. Inclusion required relevance to the manufacturing problem or a method used for RQ1–RQ4.

Summary of Key Literature

Thielen et al. (2020) examine machine-learning detection of false calls in surface-mounted technology manufacturing. Their work establishes the practical role of supervised inspection decision support and motivates RQ2 and RQ3. However, an industrial precedent does not establish that a rule trained on another line, feature representation, or evaluation unit will transfer here. The present study therefore evaluates group-level outcomes in its own recording window.

Pfab et al. (2024) describe the Siemens dataset, including scaled measurements, AOI-only selection, potential distribution changes, and unknown human label error. The article provides the context required to interpret EDA and is relevant to all four questions. Pfab's (2024) Version 2 release supplies the actual data artifact used in this project. These are related sources, not two independent empirical confirmations. Their combined relevance is provenance

and context, while the analysis must supply its own tested review policy. Table A2 summarizes the methodological role of at least ten sources.

Literature Synthesis on Model Choice

Breiman (2001) introduces random forests as ensembles of randomized trees. Their ability to model interactions motivates a bounded forest comparator for heterogeneous inspection measurements. The methodological contribution supports testing nonlinear structure, but it supplies no performance estimate for this dataset. A small positive training population also makes unrestricted trees inappropriate. Maximum depth and minimum leaf size will therefore be constrained during development.

Friedman (2001) develops gradient boosting through iterative function approximation. This supports a second nonlinear comparator that can capture patterns missed by a linear logit. Boosting can also fit unstable rare patterns, so iteration count, leaf count, and regularization must be bounded. Penalized logistic regression remains the interpretable starting model: its coefficients summarize conditional associations after the specified encoding, while nonlinear models test whether additional complexity improves chronological ranking.

Saito and Rehmsmeier (2015) compare precision–recall and receiver-operating-characteristic displays under imbalance. Their findings motivate group average precision as the development ranking metric and require reporting its prevalence reference. High overall accuracy can coexist with complete failure to capture defects when negative groups dominate. A review policy instead needs explicit sensitivity and workload accounting.

Together, these studies support comparison among interpretable and nonlinear candidates without treating algorithm sophistication as the goal. The practical gap is a group-level threshold that preserves detection under later production conditions. A better ranking score alone cannot answer that gap: the decision depends on a frozen threshold, explicit false negatives, and dependence-aware uncertainty. The present interim analysis now supplies the initial retrospective evidence while reserving deployment and external validity for later study.

Literature Synthesis on Validity

Roberts et al. (2017) examine validation strategies for structured observations, including temporal and hierarchical data. Their framework supports keeping complete days and timestamp groups together during partitioning. Random record splits could place related observations in both training and evaluation data and measure interpolation rather than the intended later-period performance. Chronological folds are therefore part of the substantive study design, not merely a software preference.

Varma and Simon (2006) show that choosing model settings using cross-validation and then reporting the same optimized error estimate can produce selection bias. Their simulation-based findings motivate separate model-selection, threshold-selection, and final-test stages. The current project will choose among candidates within training days, use validation days to freeze the threshold, and evaluate that frozen rule on the final period.

Brown et al. (2001) assess binomial interval procedures and motivate score-based estimation rather than reliance on a simple Wald interval. Their findings are relevant to defect proportions, sensitivity, and avoidance. Cohen (1992) provides effect-size and power-planning conventions used to make the RQ1 assumption explicit. Neither source removes dependence between groups sharing production conditions.

Gama et al. (2014) survey concept-drift adaptation and distinguish changing data-generating relationships from a single performance summary. This cautions against labeling the RQ4 prevalence difference as confirmed concept drift. Across the review, the main gap is operational validation that combines rare-defect uncertainty, temporal separation, and review capacity. The present interim evidence adds frozen retrospective performance and uncertainty, while prospective operational validity remains unestablished.

Data Description

Data Sources and Access

The original source is Publication data: Data of automated optical inspection of surface-mounted technology electronic production, Version 2 (Pfab, 2024), hosted at Mendeley Data under DOI 10.17632/99jzmh9658.2. The project repository contains a source manifest, download script, audit script, dictionary, and audit outputs. Data access is source-linked rather than based on an unofficial competition copy. The report's Appendix B records reproducible access instructions.

Dataset Overview

The retrieved audit reports 440,274 inspection records and 78 columns, including an export index. There are 77 substantive variables: timestamp, class, inspection_type, four metadata fields, and 70 inspection measurements. The record-level label has 4,622 positives and 435,652 negatives. Grouping by timestamp yields 39,742 groups, of which 875 are positive. These different units and denominators must remain separate throughout the analysis.

The recorded timestamps span June 23 through November 2 in the anonymized year 1970. There are 101 active production days. The publication describes a 132-day recording; active days count dates with observations and must not be confused with the full elapsed recording window. Calendar dates support ordering rather than historical interpretation.

The supplied dataset.csv and mapping.json match both checksums in the original source manifest. A fresh audit reproduces all synopsis row, group, label, and partition counts. Thus the analysis now uses verified raw observations rather than aggregate counts alone. The run records exact package versions, seed 640, fold boundaries, configuration scores, the frozen threshold, group predictions, and generated summaries. Appendix B explains how to reproduce the complete analysis. Figure A7 previews real records; Table A14 illustrates measurement summaries.

Data Dictionary

The class field is a binary manual inspection outcome, with 1 denoting a confirmed defect and 0 denoting a nondefective alarm. Inspection_type is a nominal code with five levels. The four meta_feat fields are anonymous metadata; their encoded values must not be interpreted as ordered physical measurements without semantic evidence. The inspection_feat fields are numerical measurements scaled by the provider, with applicability varying by inspection type (Pfab et al., 2024).

Timestamp is parsed as a UTC datetime for consistent grouping and partitioning. The export index is an identifier and excluded from predictors. Derived group_defect is binary; dominant_type is categorical; active_day is ordered; period identifies the early or late cohort. Group_score will be the maximum spot score within a timestamp group, and retain_review will equal one when that score meets the frozen threshold. Table A3 provides a compact dictionary; the repository dictionary enumerates raw fields individually.

Data Completeness and Limitations

The raw-file audit finds zero literal nulls, including within applicable measurement cells. Zero still does not establish measurement availability: nonapplicable fields use structural zeros. Five measurements—inspection_feat14, 15, 35, 36, and 37—are absent from mapping.json and are excluded. The remaining 65 fields generate 247 applicable feature–type pairs. Physical units remain withheld; encoded categorical measurements may be represented numerically where their semantic identity is unknown.

The dataset omits AOI-accepted products, verified board identifiers, review duration, staffing costs, and downstream outcomes. Human labels may contain errors. These limitations constrain validity more strongly than the nominal row count: factory yield, missed defects among AOI-accepted units, and realized savings cannot be calculated from this selected alarm stream.

GitHub Data Availability Statement

The analysis code, derived datasets, and reproducibility materials are available in the project repository at https://github.com/sahasrm06/Capstone_Proposal. Analysis scripts are analyze_full.py, robustness.py, and export_model.py in the repository root, with dependencies in requirements_full.txt. Derived datasets are full_results/group_data.csv and full_results/test_predictions.csv. Model-selection scores, the frozen-rule configuration, evaluation metrics, diagnostics, and figures are provided in full_results/. The complete data dictionary is docs/data_dictionary.csv. The README describes the analysis workflow and file locations.

Original raw data are accessed from the Mendeley Data Version 2 release at https://data.mendeley.com/datasets/99jzmh9658/2 (Pfab, 2024). Download dataset.csv and mapping.json and place them in data/raw/ in a local working copy. The repository's data/source_manifest.json and scripts/download_data.py document retrieval and checksum verification. Raw files are source-linked rather than redistributed, preserving the provider's attribution and noncommercial-use conditions. Figure A7 previews actual records; Figure A8 documents the accompanying analysis package. This arrangement distinguishes published derived outputs from original data held by the provider.

Analysis

Data Cleaning

The preparation workflow preserves both source files and verifies their byte sizes and SHA256 digests against the manifest. Schema, binary labels, five inspection codes, timestamp parsing, and mapping coverage pass validation. Reproduction of 440,274 rows, 39,742 groups, and 101 active days provides a second check against import or version errors. Source files are preserved unchanged and derived group tables are saved separately.

There are 8,752 repeated records excluding the export index (1.99%). The primary fit retains them because their origin is unknown. Removing exact repeats leaves 431,522 rows and preserves every group outcome, but changes dominant type for 129 groups. RQ1 is therefore repeated on the deduplicated table. At fixed fitted scores, repeated identical records do not change a group maximum; this is verified for all test groups. Retraining after deduplication remains a separate robustness task.

Applicability-aware diagnostics cover 25,590,456 applicable measurement cells. All lie within [0, 1], with no literal missing values. The 1.5-interquartile-range screen flags 2,379,205 cells (9.30%). These are distribution flags, not invalid rows: bounded, skewed, or partly categorical measurements can exceed these fences legitimately. No records are dropped or clipped on this basis. Field-level quantiles, boundary counts, and flags are saved in measurement_audit.csv.

Applicable measurements are gated by inspection type, with type indicators preserving context. Median imputation, standardization, constant removal, and categorical encoding are fitted within each training fold. Imputation is a protective pipeline step; verified inputs contain no missing cells. Unseen categories are ignored safely. No synthetic oversampling is used, and extreme values are retained as possible defect signals. Table A4 records the completed checks.

EDA Results (Interpretation Emphasis)

Configuration Risk

The alarm stream is strongly imbalanced: 4,622 of 440,274 spots are positive (1.05%), whereas 875 of 39,742 timestamp groups are positive (2.20%). A group is positive if any constituent observation is positive. Median group size is six records, with an interquartile range of 2–15 and a maximum of 557. Larger groups offer more opportunities for a high maximum spot score, so the eventual ranking is not interpreted as a calibrated group probability.

Figure A1 and Table A5 show that dominant type 1 has the highest observed group defect proportion, $\frac{197}{3{,}404}$ or 5.79%. Types 0 and 2 have proportions of approximately 1.85% and 1.76%, respectively. Type 4 contributes only 269 groups and six positives, so its 2.23% estimate has substantially less precision. High prevalence is not equivalent to the largest contribution to total defects: type 2 contains 243 positive groups because its overall volume is much larger.

The omnibus Pearson statistic is $\chi^2(4) = 224.145$, $p < .001$, with Cramer's $V = 0.0751$. The smallest expected positive count is 5.92, satisfying the usual cell-size check for this full table. RQ1 therefore has preliminary evidence of association under independent-group assumptions. The small effect size and unequal volumes matter operationally: type alone should not become an automatic bypass rule. Day-clustered inference and composition checks remain necessary, and no causal explanation is assigned to anonymous type codes.

Temporal Variation

The chronological training partition contains 14,975 groups and 179 positives, giving prevalence 1.20%. Validation contains 5,627 groups and 82 positives, or 1.46%. The final test contains 19,140 groups and 614 positives, or 3.21%. Figure A2 displays these proportions. The later test population is substantially different in outcome frequency from training, which makes a random split particularly unsuitable for the intended later-period evaluation.

RQ4 uses a different, prespecified period definition: the first 50 versus last 51 active days. Early prevalence is $\frac{215}{17{,}972}$ = 1.20%, and late prevalence is $\frac{660}{21{,}770}$ = 3.03%. Figure A3 shows the absolute difference of 1.835 percentage points; the late-to-early prevalence ratio is approximately 2.53. The independent-group pooled score test gives $Z = 12.410$, $p < .001$. The unpooled 98.75% interval for the difference is [1.481, 2.189] percentage points.

These results answer the descriptive part of RQ4: recorded periods differ in confirmed-defect prevalence. A day-clustered logistic model adjusting for dominant type gives period odds ratio 3.10, with $p < .001$. The three-day block 98.75% interval for the unadjusted prevalence difference is [0.326, 3.645] percentage points. Figure A4 displays daily variation. Inspection mix and unknown operator or process changes may contribute; neither the period effect nor a daily spike identifies a causal change or proves concept drift. Table A16 consolidates the principal EDA insights and their implications for the four research questions.

Sample Size Calculations

Bonferroni allocation of familywise alpha = .05 gives alpha = .0125 per question. Critical normal values are 2.497705 and .841621 for 80% power. Table A6 compares requirements and available counts. Planning precision and power do not promise successful policy performance.

RQ1 uses a 2 × 5 table with four degrees of freedom, Cohen's w = .10, alpha = .0125, and 80% power. The noncentral chi-square calculation is (Cohen, 1992):

$$\lambda = Nw^2 = 0.01N,\ c = \chi^2_{4,0.9875} = 12.761851$$

$$Pr[\chi^2_4(0.01N) > 12.761851] \geq 0.80 \ \Rightarrow \ N_1 = 1{,}611$$

RQ2 plans sensitivity precision at p = .90 and half width d = .05. The requirement is 225 positive groups; the cohort conversion uses overall prevalence π = $\frac{875}{39{,}742}$:

$$m_2 = \left\lceil \frac{z^2 p(1-p)}{d^2} \right\rceil = \left\lceil \frac{2.497705^2(0.90)(0.10)}{0.05^2} \right\rceil = 225$$

$$N_2 = \left\lceil \frac{225}{\frac{875}{39{,}742}} \right\rceil = 10{,}220$$

The actual final test contains 614 positive groups, exceeding the event requirement. Cohort conversion is a projection, not a guarantee of positive counts.

RQ3 uses p = .50 and d = .02. For RQ4, p_1 = .02, p_2 = .03, $\bar{p}$ = .025, and power is 80% with equal allocation:

$$N_3 = \left\lceil \frac{2.497705^2(0.50)(0.50)}{0.02^2} \right\rceil = 3{,}900$$

$$n = \left\lceil \frac{\left[z\sqrt{2\bar{p}(1-\bar{p})} + z_{0.80}\sqrt{p_1(1-p_1) + p_2(1-p_2)}\right]^2}{(p_2 - p_1)^2} \right\rceil$$

$$n = 5{,}435\ per\ period; \quad N_4 = 2n = 10{,}870$$

$$N_{min} = max\{1{,}611;\ 10{,}220;\ 3{,}900;\ 10{,}870\} = 10{,}870$$

All groups are retained. Event and allocation constraints remain separate. A design effect of two is a sensitivity assumption; stronger dependence may yield inconclusive findings.

Modelling

Choice of Models with Justification

The operational baseline retains every group for manual review. Penalized logistic regression, bounded random forest, and histogram gradient boosting were compared across the approved 14-candidate grid (Breiman, 2001; Friedman, 2001). Model selection used only the three expanding training-day folds. The selected histogram gradient boosting has mean development group average precision 0.4568; Table A9 gives all candidates, folds, and parameters. No final-test result was used to choose a model.

Model 1 Penalized Logistic Regression

This interpretable linear starting model limits complexity through regularization. It estimates conditional associations after type gating and encoding, but cannot represent arbitrary nonlinear interactions without additional terms.

Model 2 Random Forest

Bounded forests test nonlinear interactions in the tabular AOI inputs. Depth and minimum-leaf constraints reduce rare-pattern fitting; individual split importance remains descriptive and potentially biased (Breiman, 2001).

Model 3 Histogram Gradient Boosting

Regularized boosting tests additive nonlinear structure with a limited leaf budget. The winning specification uses 100 iterations, 15 leaves, and L2 = 10, selected solely by training-period folds (Friedman, 2001).

Features Included and Feature Engineering

Models train on spot-level records and aggregate maximum spot scores to groups. This maximum is a ranking score, not a calibrated group-defect probability. The 247 gated measurement–type pairs and one-hot inspection type and metadata form the inputs. The source

article describes metadata as AOI/context fields available before manual review (Pfab et al., 2024). Raw timestamp, export index, class, and all label-derived summaries are excluded. Unknown encoded measurement semantics remain a limitation.

Validation and Threshold Selection

Within the first 40 active days, expanding folds train on days 1–20, 1–25, and 1–30 and assess days 21–25, 26–30, and 31–40. Each fold refits preprocessing. Candidate ranking uses mean group average precision, with exact ties favoring the earlier, simpler candidate. The grid includes six logistic specifications, four forests of 200 trees, and four boosting specifications. Data and algorithms use seed 640 and one CPU worker.

The separate validation partition selects the threshold only after the winner is refitted on training days. The frozen threshold is 0.0015671194. Validation sensitivity is 98.78%, its 98.75% Wilson lower endpoint is 90.81%, and avoidance is 17.63%. The rule is not refitted on validation data. Final-day outcomes are evaluated once for the primary result; subsequent summaries assess that fixed rule (Roberts et al., 2017; Varma & Simon, 2006).

Evaluation Metrics (Include Formulae and Calculations)

For a fixed group rule, TP counts positive groups retained; FN counts positive groups assigned below threshold; FP counts negative groups retained; and TN counts negative groups assigned below threshold. The policy metrics are:

$$Sensitivity = \frac{TP}{TP + FN} \quad Precision = \frac{TP}{TP + FP}$$

$$Avoidance = \frac{TN + FN}{N} \quad Missed - defect\ rate = \frac{FN}{TP + FN}$$

$$Confirmed - negative\ avoidance = \frac{TN}{N}$$

Avoidance includes missed positive groups; it must be distinguished from confirmed-negative avoidance. Table A17 provides worked results, and Tables A10–A11 report uncertainty. Appendix C collects the equations.

Preliminary Results

Preliminary Model Performance

The review-all baseline on the audited test groups has TP = 614, FN = 0, FP = 18,526, and TN = 0. Sensitivity is 100%, precision is 3.21%, avoidance is 0%, and missed-defect rate is 0%. The 98.75% Wilson interval is [98.994%, 100%] for sensitivity and [0%, 0.0326%] for avoidance (Brown et al., 2001). This baseline clears the independent-group RQ2 detection criterion but fails RQ3. It is an arithmetic policy evaluation, not a trained model experiment.

The learned rule yields TP = 604, FN = 10, FP = 16,340, and TN = 2,186. Sensitivity is 98.37% and candidate avoidance is 11.47%; Table A10 provides intervals and denominators. Group average precision is 0.1299, against the .0321 constant-score reference. Figure A5 displays precision–recall and Figure A6 shows the frozen operating point on the retrospective tradeoff curve. The test curve is descriptive and is not used for retuning (Saito & Rehmsmeier, 2015). Table A12 reports subgroup counts; Table A13 maps conclusions to each RQ.

Preliminary Findings by Research Question

RQ1 Configuration Risk

The raw group contingency table indicates association, $\chi^2(4) = 224.145$, with small Cramer's $V = .0751$. The day-clustered omnibus $p = .00114$ also falls below .0125. Duplicate removal preserves a similar association. Interpretation remains descriptive; physical configuration causes cannot be identified from anonymous codes. See Tables A10–A13 for counts and uncertainty.

RQ2 Defect Capture

The fixed rule captures $\frac{604}{614}$ positive test groups (98.37%), with ten missed positive groups. The 98.75% Wilson lower endpoint is 96.53%; three-day block sensitivity has lower endpoint 96.21%. The retrospective detection criterion is supported for this recording, while

prospective operational validation remains pending. See Tables A10–A13 for counts and uncertainty.

RQ3 Candidate Review Avoidance

The rule assigns $\frac{2,196}{19,140}$ groups below threshold (11.47%), including 2,186 negatives and ten positives. The Wilson lower endpoint is 10.91%, but the three-day block lower endpoint is 7.79%. Consequently, the joint dependence-aware avoidance claim is inconclusive; no actual reviews are removed. See Tables A10–A13 for counts and uncertainty.

RQ4 Temporal Prevalence

Early and late prevalence are 1.20% and 3.03%, differing by 1.835 percentage points. The three-day block difference interval is .326–3.645 percentage points, and adjusted period $p < .001$. Temporal variation is supported; causal process change and concept drift are not established. See Tables A10–A13 for counts and uncertainty.

Interim Limitations and Risks

The principal limitations are single-line sampling, anonymous measurement semantics, uncertain human labels, and timestamp groups as a board proxy. All analyses remain conditional on the AOI alarm stream. Row count cannot repair these limits. The full development grid and frozen evaluation are completed, but independent external validation, deduplicated-model retraining, and prospective review-time measurements remain future work.

Day dependence is assessed using 2,000 circular moving-block resamples at block lengths one, three, and seven active days. Models and thresholds remain fixed. At three days, sensitivity has a 98.75% percentile interval [96.21%, 100.00%] and avoidance [7.79%, 15.75%]. Table A11 reports all block lengths. These are conditional bootstrap sensitivity estimates, not guarantees of exact simultaneous coverage or training-sample stability.

Next Steps for the Final Report

The next phase will address the remaining validity and submission tasks:

- Maintain repository documentation and preserve the published interim configuration for independent reproduction.
- Reproduce the pinned-environment workflow independently and compare all saved counts and metrics.
- Verify feature semantics and repeat model development after duplicate removal; use a new evaluation period for any model revision.
- Assess subgroup stability and prospective review duration; retain manual review during any shadow evaluation.

The joint dependence-aware review-avoidance conclusion remains inconclusive because its three-day lower endpoint is 7.79%, below 10%. Retain manual inspection; scores may support queue prioritization. No realized saving or deployment benefit is established.

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Appendix A

Supporting Tables and Figures

Table A1

Research Question Evidence Matrix

Question	Interim evidence	Remaining work
RQ1	Raw table and clustered sensitivity	Interpret association, not causation
RQ2	Frozen test sensitivity and intervals	Prospective confirmation
RQ3	Frozen test avoidance conditional on RQ2	Joint decision below
RQ4	Period test, adjusted model and blocks	External time window

Table A2

Literature Relevance Matrix

Source	Method and contribution	Application
Thielen et al. (2020)	Manufacturing false-call classification	RQ2 and RQ3 motivation
Pfab et al. (2024)	Real production data description	Context and limitations
Pfab (2024)	Versioned data release	Reproducible input
Breiman (2001)	Randomized tree ensembles	Nonlinear comparator
Friedman (2001)	Additive boosting	Nonlinear comparator
Saito and Rehmsmeier (2015)	Imbalanced metric comparison	Average precision
Roberts et al. (2017)	Structured validation review	Chronological partitions
Varma and Simon (2006)	Selection-bias simulations	Separate tuning and testing
Brown et al. (2001)	Binomial interval comparison	Wilson uncertainty
Cohen (1992)	Effect-size power planning	RQ1 feasibility
Gama et al. (2014)	Concept-drift survey	RQ4 interpretation

Table A3

Data Dictionary Summary

Variable family	Type	Meaning and role
timestamp	Datetime	Anonymized time; grouping and split
class	Binary integer	Manual outcome; target only
inspection_type	Nominal integer	Codes 0–4; configuration
meta_feat1–4	Encoded metadata	Anonymous categories; semantic review needed
70 inspection_feat fields	Numeric	Scaled AOI inputs; type applicability
Unnamed: 0	Integer ID	Export index; excluded
group_defect	Binary derived	Maximum class per timestamp
dominant_type	Nominal derived	Mode type; smallest-code tie
active_day and period	Ordered and binary	Chronology and RQ4
group_score and retain_review	Numeric and binary	Future score and policy output

Note. Individual raw variable names and applicability are enumerated in project/docs/data_dictionary.csv in the supporting package. Gaps in measurement numbering are preserved.

Table A4

Cleaning and Audit Status

Check	Evidence	Action
Literal nulls	0 in raw and applicable cells	Protective imputation only
Repeated rows	8,752; 1.99%	Retained; group sensitivity checked
Unmapped fields	5 of 70 measurements	Excluded; 247 gated inputs
Outliers	2,379,205 cell IQR flags	Retained; not invalid-row count
Source hashes	Both match source manifest	Verified before analysis
Partitions	40 / 20 / 41 active days	Disjoint groups and days

Table A5

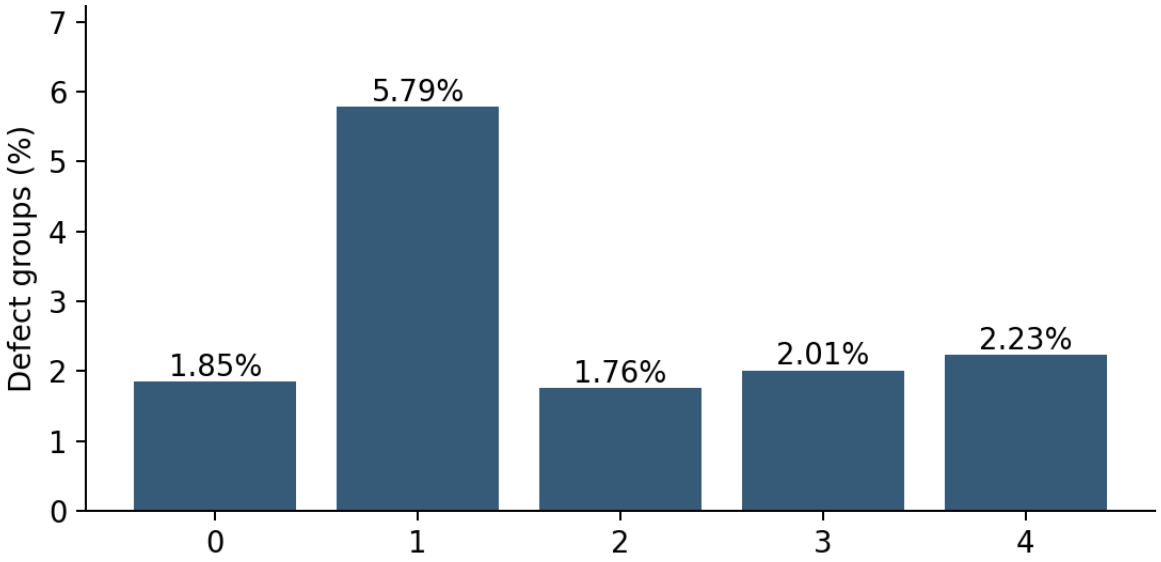
Dominant Inspection Type and Confirmed Defects

Type	Groups	Positive	Defect %
0	12037	223	1.853
1	3404	197	5.787
2	13787	243	1.763
3	10245	206	2.011
4	269	6	2.230

Note. Pearson $\chi^2(4) = 224.145$; $p < .001$; Cramer's $V = 0.0751$. Group independence is an unverified assumption.

Figure A1

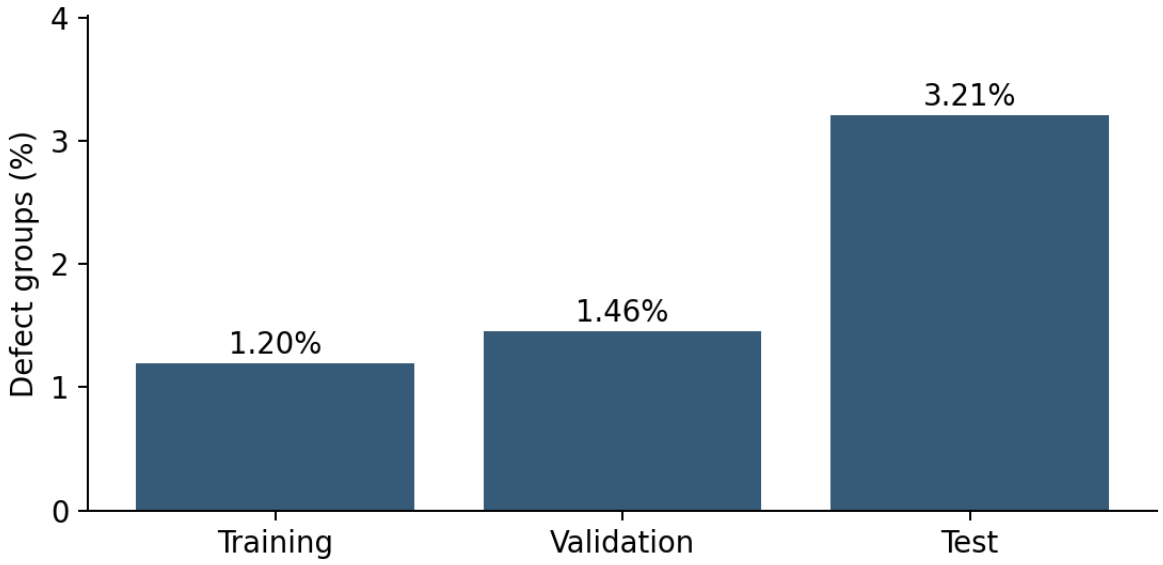
Defect Prevalence by Dominant Inspection Type



Note. Type 1 has the highest observed risk; type 4 has only six positive groups. Source: repository audit and dominant-type counts; calculations for this report.

Figure A2

Defect Prevalence Across Chronological Partitions

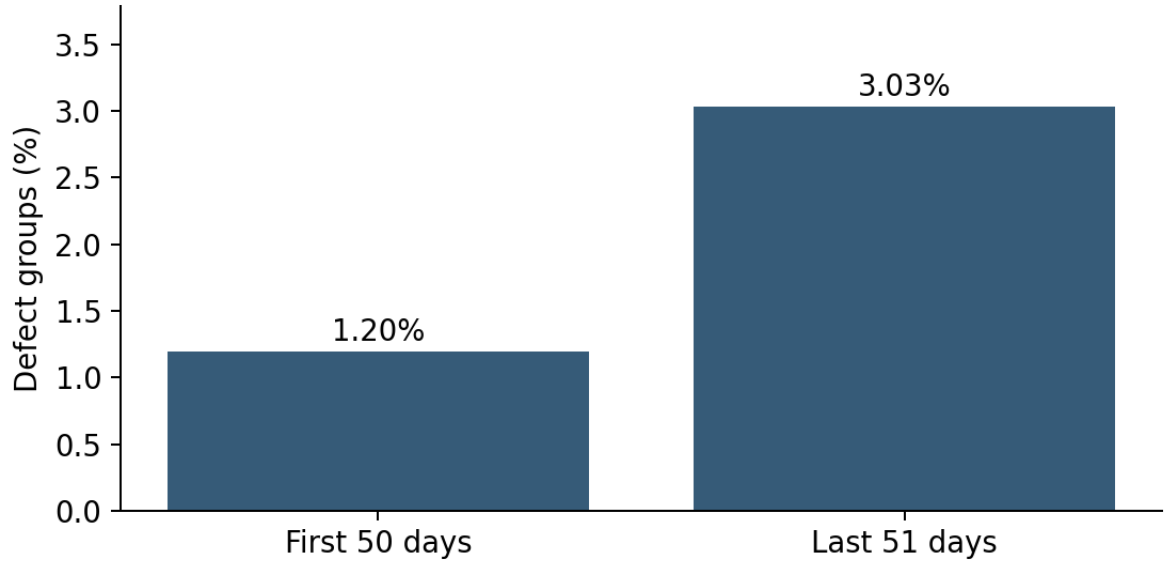


Note. Later test prevalence exceeds development prevalence; partitions are not exchangeable.

Source: repository audit and dominant-type counts; calculations for this report.

Figure A3

Early and Late Period Defect Prevalence



Note. Period variation is descriptive and does not identify its cause. Source: repository audit and dominant-type counts; calculations for this report.

Table A6

Sample Requirements and Available Observations

RQ	Planning requirement	Available
1	1,611 groups	39,742
2	225 positive groups	614 test positives
3	3,900 test groups	19,140
4	5,435 groups each period	17,972 early; 21,770 late

Note. Alpha = .0125 per question. RQ1 and RQ4 use 80% power; RQ2 and RQ3 plan interval precision. Cohort equivalents do not replace positive-event constraints.

Table A7

Feature Set and Model Usage

Input	Transformation	Leakage control
Mapped AOI fields	Type gating; training scaling	Fit within training folds
Type and eligible metadata	One-hot encoding	Ignore unseen categories
Index and timestamp	Excluded from predictors	Used only for audit and splits
Manual class	Outcome only	Never included as input
Group maximum score	Aggregate spot predictions	Threshold selected on validation

Table A8

Preliminary Policy Performance

Policy	Sensitivity	Precision	Avoidance
Review all	$\frac{614}{614}$ = 100%	$\frac{614}{19{,}140}$ = 3.21%	$\frac{0}{19{,}140}$ = 0%
Logistic regression	Development grid; see A9	Development grid; see A9	Development grid; see A9
Random forest	Development grid; see A9	Development grid; see A9	Development grid; see A9
Gradient boosting	Development grid; see A9	Development grid; see A9	Development grid; see A9

Note. Review-all TP = 614, FN = 0, FP = 18,526, TN = 0. Its accuracy is 3.21%. Learned final-test metrics are in Table A10.

Supplementary Model and Robustness Results

Table A9

Chronological Development Model Comparison

ID and family	Parameters	Fold AP values	Mean AP
0 LR	C=0.01; weight=None	0.4021, 0.3058, 0.0936	0.2672
1 LR	C=0.01; weight=balanced	0.2815, 0.3451, 0.1222	0.2496
2 LR	C=0.1; weight=None	0.2661, 0.2920, 0.0864	0.2148
3 LR	C=0.1; weight=balanced	0.2690, 0.3373, 0.0829	0.2297
4 LR	C=1; weight=None	0.3502, 0.1577, 0.0811	0.1964
5 LR	C=1; weight=balanced	0.2404, 0.3335, 0.0406	0.2048
6 RF	depth=3; leaf=50	0.3501, 0.2870, 0.0493	0.2288
7 RF	depth=3; leaf=200	0.4220, 0.1953, 0.0503	0.2225
8 RF	depth=6; leaf=50	0.3641, 0.3513, 0.0508	0.2554
9 RF	depth=6; leaf=200	0.4373, 0.3381, 0.0479	0.2745
10 HGB	leaves=7; L2=1	0.6446, 0.5165, 0.1206	0.4272
11 HGB	leaves=7; L2=10	0.5362, 0.5137, 0.1419	0.3973
12 HGB	leaves=15; L2=1	0.5949, 0.5459, 0.1448	0.4285
13 HGB	leaves=15; L2=10	0.6262, 0.5891, 0.1551	0.4568

Note. AP = group average precision. Folds evaluate days 21–25, 26–30, and 31–40. LR = logistic regression; RF = random forest; HGB = histogram gradient boosting. Only training-period folds select the model. Ordinary class weights are denoted None. AP values are not final-test performance estimates.

Frozen Policy Evaluation

Table A10

Final Test Group Performance

Metric	Estimate and denominator	98.75% Wilson interval
Sensitivity	$\frac{604}{614}$ = 98.371%	96.525% to 99.244%
Avoidance	$\frac{2196}{19{,}140}$ = 11.473%	10.910% to 12.061%
Precision	$\frac{604}{16944}$ = 3.565%	Secondary descriptive metric
Confirmed-negative avoidance	$\frac{2186}{19{,}140}$ = 11.421%	Subset of total avoidance
Missed positive groups	$\frac{10}{614}$ = 1.629%	Complement of sensitivity
Group average precision	0.12986	Constant-score reference .03208

Note. Both independent-group criteria pass; block sensitivity checks do not establish the joint targets. TP = 604; FN = 10; FP = 16340; TN = 2186. Confusion counts sum to 19,140. The selected model is histogram gradient boosting; threshold 0.0015671194. No threshold was chosen using these test outcomes.

Table A11

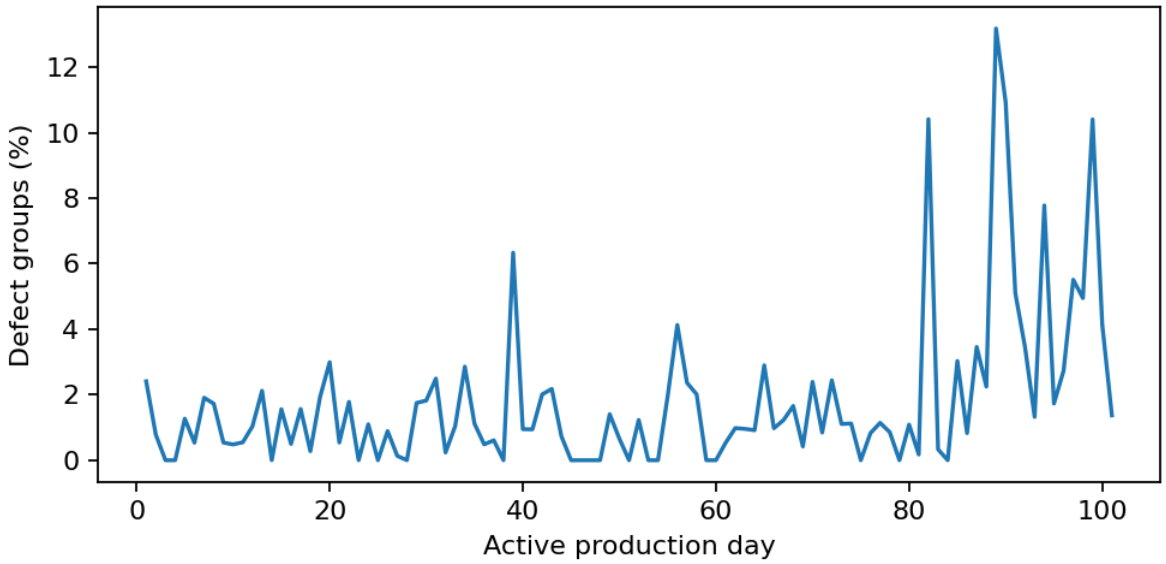
Dependence Sensitivity at Fixed Fitted Rule

Block length	Sensitivity interval %	Avoidance interval %	Period difference pp
1	96.17 to 100.00	7.72 to 15.83	0.527 to 3.227
3	96.21 to 100.00	7.79 to 15.75	0.326 to 3.645
7	96.75 to 100.00	8.42 to 14.68	0.161 to 3.798

Note. 2,000 circular moving-block resamples; 98.75% percentile intervals. Block length is in active days. pp = percentage points. RQ4 blocks are sampled separately within each period. Fixed model and threshold intervals do not include model-selection uncertainty.

Figure A4

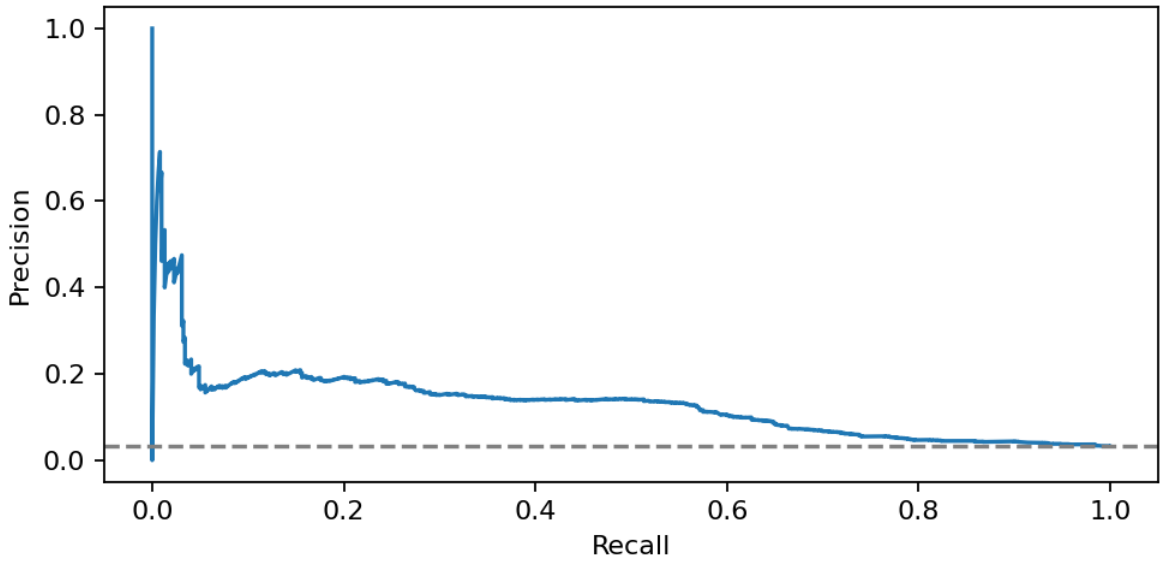
Daily Confirmed Defect Prevalence



Note. Denominators vary by active day. Peaks indicate local concentration, not an identified causal process change.

Figure A5

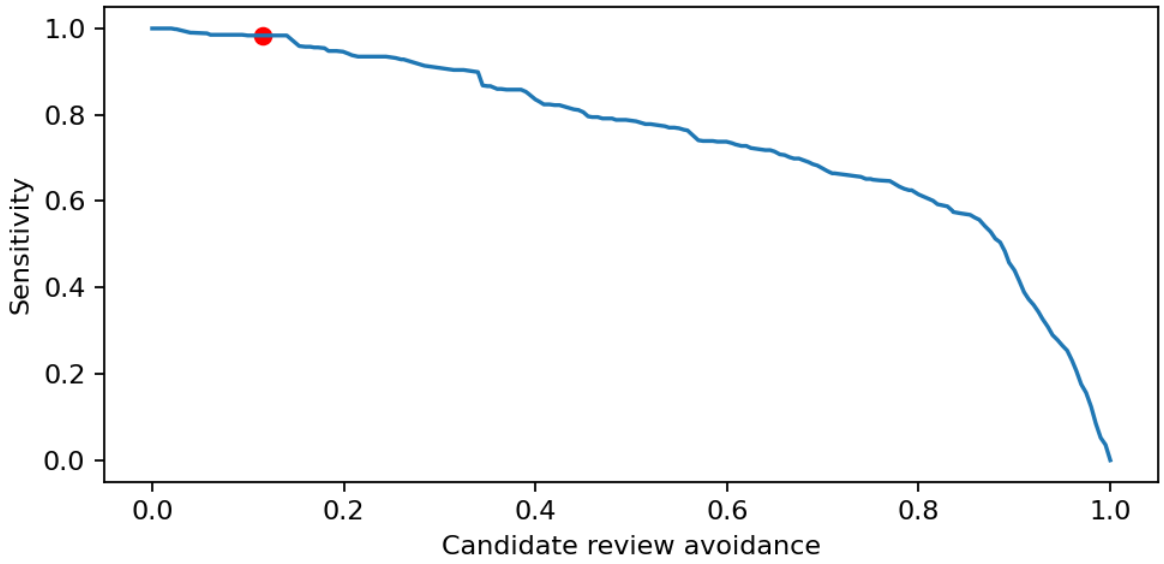
Frozen Model Precision and Recall on Final Test Groups



Note. Dashed line marks the test-prevalence reference. This display describes final scores and was not used for tuning.

Figure A6

Retrospective Detection and Review Avoidance Tradeoff



Note. Red point marks the validation-frozen operating threshold. Other points are descriptive only and cannot justify a revised threshold on this same test.

Subgroup Diagnostics and Research Decisions

The day-clustered dominant-type omnibus Wald statistic is 18.183 on four degrees of freedom, $p = 0.001137$. The adjusted late-period log odds coefficient is 1.1307, cluster standard error 0.2658, $p < .001$. These are sensitivity associations rather than causal process effects.

Table A12

Frozen Test Results by Dominant Type

Type	Groups	Positive	Captured	Avoidance %
0	6667	185	177	14.20
1	1118	150	150	0.54
2	6039	141	140	14.27
3	5197	134	133	7.33
4	119	4	4	0.00

Note. Small positive counts make some subgroup detection estimates unstable. These slices are diagnostic and do not select a threshold.

Table A13

Interim Answers to the Four Research Questions

RQ	Interim answer
1	Configuration association: independent χ^2 $p < .0125$; day-cluster $p = 0.001137$.
2	Frozen sensitivity 98.37%; Wilson lower 96.53%. Criterion met.
3	Avoidance 11.47%; Wilson lower 10.91%. Joint independent criteria met.
4	Late prevalence higher by 1.835 pp; adjusted cluster $p < .001$. Three-day interval 0.326 to 3.645 pp.

Note. Both independent-group criteria pass; block sensitivity checks do not establish the joint targets. Secondary slices do not change the four primary questions or justify retuning.

Screenshots and Evidence

Figure A7

Preview of Five Real Source Records and Selected Columns

Timestamp	Class	Type	Meta 1	Feature 1
1970-06-23 03:58:55+00:00	0	0	0	0.508
1970-06-23 03:58:55+00:00	0	0	12	0.508
1970-06-23 03:59:34+00:00	0	0	8	0.508
1970-06-23 03:59:34+00:00	0	0	0	0.508
1970-06-23 03:59:34+00:00	0	0	8	0.508

Note. Preview uses original records. Full field names, applicability, and types are provided in the supporting dictionary. Timestamp year is anonymized.

Table A14

Example Applicable Measurement Summaries

Field	Applicable *n*	Median	Q1 to Q3	IQR flags
inspection_feat1	440274	0.508	0.500 to 0.540	67786
inspection_feat2	440274	0.181	0.181 to 0.181	62634
inspection_feat3	440274	0.479	0.479 to 0.499	110885
inspection_feat4	440274	0.434	0.433 to 0.434	207029
inspection_feat5	440274	0.500	0.500 to 0.500	135202
inspection_feat6	440274	0.395	0.395 to 0.395	71853
inspection_feat7	440274	0.351	0.351 to 0.351	78784

Note. Summaries pool applicable types for compact screening only; model inputs retain separate type contexts. Flags are cell-level distribution indicators. The complete CSV provides all 65 mapped fields; no outlier deletion was applied.

Duplicate and Feature Importance Sensitivity

Removing 8,752 exact repeats preserves all group outcomes and frozen test group scores, but changes dominant type in 129 groups. Deduplicated RQ1 chi-square is 226.559, $p < .001$, $V = 0.0755$. Association remains similar. Retraining after deduplication remains future work.

Table A15

Leading Fitted Feature Contributions

Encoded feature	Relative contribution
cat__meta_feat1_27	0.2222
n__inspection_feat96__type2	0.0824
n__inspection_feat48__type1	0.0767
n__inspection_feat94__type3	0.0554
n__inspection_feat12__type3	0.0357
cat__meta_feat1_3	0.0317
n__inspection_feat96__type3	0.0302
cat__meta_feat4_4	0.0299
n__inspection_feat95__type2	0.0227
cat__meta_feat4_16	0.0217

Note. Normalized positive training split gains from the pinned histogram booster. Correlated inputs share contributions, and flexible fields may be favored. Anonymous importance is descriptive, not causal, and did not change the threshold.

Metadata-only validation average precision is 0.0248 for logistic C = .1. This secondary diagnostic did not select the rule. Deterministic refitting reproduced every saved primary test score within numerical tolerance.

Additional Template Tables

Table A16

EDA Insight Summary

Evidence	Interpretation	Research implication
$\frac{875}{39{,}742}$ positive groups; 2.20%	Rare outcomes make accuracy misleading	RQ2 requires event-based sensitivity; RQ3 needs missed-defect counts
Median group size 6; maximum 557	Maximum score depends partly on group opportunity count	Report group units and avoid probability calibration claims
Type 1 prevalence 5.79%; type 2 has 243 positive groups	Risk concentration and workload volume differ	RQ1 supports configuration diagnostics, not a type-only bypass
Training, validation, and test prevalence: 1.20%, 1.46%, and 3.21%	Later outcome frequency differs	Use chronological validation for RQ2/RQ3
Late minus early prevalence 1.835 pp	Temporal prevalence variation is supported	RQ4 needs dependence-aware intervals and mix adjustment

Table A17

Evaluation Metrics and Worked Calculations

Metric	Formula and result	Purpose
Sensitivity	$S = \frac{TP}{TP + FN}$	RQ2; denominator is positive groups
	$\frac{604}{614} = 0.983713$	
Precision	$P = \frac{TP}{TP + FP}$	Secondary retained-queue concentration
	$\frac{604}{16{,}944} = 0.035647$	
Avoidance	$A = \frac{TN + FN}{N}$	RQ3; includes ten missed positive groups
	$\frac{2{,}196}{19{,}140} = 0.114734$	
Confirmed-negative avoidance	$\frac{TN}{N}$	Separate negative capacity opportunity
	$\frac{2{,}186}{19{,}140} = 0.114211$	
Missed-defect rate	$\frac{FN}{TP + FN}$	Detection loss; complement of sensitivity
	$\frac{10}{614} = 0.016287$	
Average precision	$AP = \sum_k (R_k - R_{k-1}) P_k$	Ranking; compare constant-score reference .032079
	$AP = 0.129857$	

Note. EDA = exploratory data analysis; pp = percentage points. Tables A10–A11 report intervals; Appendix C displays equations.

Figure A8

Folder Listing of the Supplied Analysis Package

Package folder	Immediate entries
full_results/	24
project/	7
project/data/	2
project/docs/	2
project/results/	3
project/scripts/	3

Note. This listing is generated from the actual analysis-package directories. Original raw inputs are intentionally absent and must be placed locally in project/data/raw/ before package-based reproduction. Repository paths and the package's nested project/ preparation snapshot are distinguished in Appendix B.

Appendix B

Reproducibility and Data Availability

The accompanying supporting package contains scripts/analyze_full.py, the original preparation scripts, the source manifest and complete raw-field dictionary, group-level derived data, measurement diagnostics, model-selection scores, frozen rule configuration, test predictions, figures, and exact package versions. From the package root, place the original files under project/data/raw/, install requirements_full.txt, run the audit and sample-size scripts, then run analyze_full.py. The complete grid is CPU-intensive and prints candidate progress to the console.

Run python project/scripts/audit_data.py and python project/scripts/sample_sizes.py before the main analysis. Inputs must match project/data/source_manifest.json. The package retains a project/ snapshot of the preparation repository; in the GitHub repository itself the corresponding paths are scripts/ and data/. The raw ZIP is not redistributed. If automated source retrieval fails, download dataset.csv and mapping.json manually from Mendeley. The analysis scripts and derived outputs are also provided in the project GitHub repository.

The source release includes noncommercial-use conditions. Preserve provider attribution and notices, and do not assume permission to redistribute raw files. If the course requires raw and processed files directly in GitHub rather than source-linked access, resolve that requirement with the mentor before submission. The original public data remain accessible through the cited Mendeley release. The package also includes the complete existing raw-field dictionary.

Independent-group tests are supplemented by day-cluster logistic covariance and fixed-rule moving-block intervals. Period blocks remain separate. Duplicate checks preserve group outcomes and frozen scores; they do not replace deduplicated-model retraining.

Appendix C

Equations and Worked Calculations

Sample planning for RQ2 and RQ3

$$n = \left\lceil \frac{z^2 p(1 - p)}{d^2} \right\rceil$$

Two period sample planning for RQ4

$$n = \left\lceil \frac{\left[z\sqrt{2\bar{p}(1 - \bar{p})} + z_{0.80}\sqrt{p_1(1 - p_1) + p_2(1 - p_2)} \right]^2}{(p_2 - p_1)^2} \right\rceil$$

Wilson score interval

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1 - \hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}$$

Detection and workload

$$S = \frac{TP}{TP + FN} \quad A = \frac{TN + FN}{N} \quad P = \frac{TP}{TP + FP}$$

For all intervals, z = 2.497705. RQ2 requires 225 positive groups; RQ3 requires 3,900 groups. RQ4 uses $p_1 = .02$, $p_2 = .03$, and $\bar{p} = .025$ to obtain 5,435 per period. RQ1 requires 1,611 groups through noncentral chi-square power. These planning calculations do not promise that the fitted policy will meet the research thresholds.

Worked learned-rule calculations and the unadjusted period difference are:

$$S = \frac{604}{604 + 10} = 0.983713; \quad A = \frac{2{,}186 + 10}{19{,}140} = 0.114734$$

$$P = \frac{604}{604 + 16{,}340} = 0.035647; \quad \Delta p = \frac{660}{21{,}770} - \frac{215}{17{,}972} = 0.01835390$$

Appendix D

Template Coverage Map

Table D1 maps each template requirement to its report location. Original raw data remain source-linked through Mendeley; analysis code and derived outputs are identified in the GitHub Data Availability Statement.

Table D1

Template to Report Mapping

Template requirement	Report location	Status
Title page	Title, student, institution, course, mentor, term, date	Covered
GitHub link after title	Repository and Report Navigation	Covered
Introduction / Background and Context	Introduction / Background and Context	Covered
Problem Statement	Problem Statement	Covered
Purpose of the Study	Purpose of the Study	Covered
Interim Project Status	Progress Snapshot with Completed/In progress/Pending bullets	Covered
Scope and objectives / RQ1–RQ4	Scope and Objectives / four explicit RQ subheadings	Covered
Literature survey / review approach	Literature Survey / Literature Review Approach	Covered
Key literature and relevance matrix	Summary of Key Literature; thematic synthesis; Table A2	Covered; 11 sources including data release
Data source and access	Data Description / Data Sources and Access	Covered
Dataset overview	Dataset Overview; 440,274 rows, 78 columns, time and units	Covered
Mandatory dictionary	Data Dictionary; Table A3; complete dictionary in analysis package	Covered
GitHub data availability	GitHub Data Availability Statement; Appendix B	Covered; original raw data source-linked
Screenshots / folder evidence	Screenshots and Evidence; Figures A7 and A8	Covered

Template requirement	Report location	Status
Analysis / Data Cleaning	Analysis / Data Cleaning; Table A4	Covered
EDA results / insight summary	EDA Results; Temporal Variation; Table A16; Figures A1–A4	Covered
Modelling / models with justification	Modelling; Model 1, Model 2 and Model 3 subsections	Covered
Feature set / engineering	Features Included and Feature Engineering; Table A7	Covered
Metrics / formulae / calculations	Evaluation Metrics; Table A17; Appendix C	Covered
Preliminary results by RQ	Preliminary Findings by Research Question / RQ1–RQ4	Covered
Preliminary model comparison	Preliminary Model Performance; Tables A8–A10	Covered; development versus test distinguished
Limitations and risks	Interim Limitations and Risks; dependence and data caveats	Covered
Next steps	Next Steps for the Final Report / action bullets	Covered
Bibliography	References / 11 APA entries with in-text citations	Covered
Appendices / reproducible code	Appendices A–D and supporting analysis ZIP	Covered
Sample size rubric requirement	Sample Size Calculations; Table A6; Appendix C	Covered for all four RQs